

Net-opt notes

Me

Today

Contents

1	Standard questions	2
1.1	Critical path with time windows	2
1.2	Bipartite graph (Transportation problem)	2
1.3	Maximum flow problem	2
1.4	Prove 3 statements	2
1.5	Shortest path	3
1.6	A* algorithm for shortest path	3
1.7	Minimum cost flow problem	3
1.8	Modeling a hub	3
1.9	Hungarian Algorithm	3
1.10	Repositioning of empty containers	3
1.11	Commodities	3
1.12	Dynamic networks	3
1.13	Constructing a network	4
1.14	Resource constrained shortest path	4

1 Standard questions

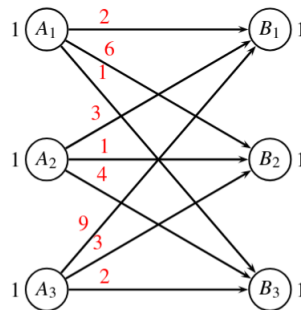
1.1 Critical path with time windows

Use forward and backward procedure to get the critical path and the makespan T

- Exam 2024 p. 1
- Exam 2023 p. 1
- Exam 2022 p. 1 - Where we have to build the graph with predecessors

1.2 Bipartite graph (Transportation problem)

When we have one side with supply and one side with demand.



- Exam 2024 p. 2 - Assignment problem
- Exam 2023 p. 2 - With more demand than supply
- Exam 2022 p. 9 - With more supply than demand.

1.3 Maximum flow problem

When we have to find the maximum flow between 2 nodes.

- Exam 2024 p. 4 - Push relabel
- Exam 2024 p. 13 - With lower bounds on edges
- Exam 2023 p. 5 - With demand and lower bounds
- Exam 2022 p. 2 - Push relabel / preflow-push

1.4 Prove 3 statements

Where we are given a problem and need to prove statements by drawing graphs

- Exam 2022 p. 4 - Max flow and min cut
- Exam 2021 p. 5 - Max flow and min cut

1.5 Shortest path

Dijkstra, bellman-ford, prim and kruskall

- Exam 2023 p. 4 - Dijkstra shortest path containing most edges.
- Exam 2020 p. 1 - Bellmann ford
- Exam 2019 p. 1 - Transform st. any algorithm would work

1.6 A* algorithm for shortest path

Dijkstra type for finding the path to a specific node

- Exam 2024 p. 6 - Manhattan distance

1.7 Minimum cost flow problem

When the objective is to minimize the cost

- Exam 2024 p. 8 - Augmenting cycle algorithm

1.8 Modeling a hub

When we have to create a model of hubs like the assignment.

- Exam 2024 p. 11 - Write values on graph based on cappcity, shipment cost and time table (Vessels arrive on diffent dates)

1.9 Hungarian Algorithm

- Exam 2023 p. 7 - Creating a bigraph from the algorithm

1.10 Repositioning of empty containers

Rewrite inventory to demand

- Exam 2023 p. 8 - Rewrite MEDITLINE from inventory to demand, to move empty containers.
- Exam 2022 p. 5 - With network simplex and augmenting cycle
- Exam 2022 p. 7 - With lower bounds where we just subtract from upper bounds.

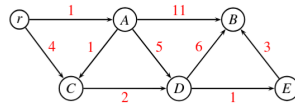
1.11 Commodities

When we have multiple commodities and find different routes.

- Exam 2023 p. 11 - The normal form of the problem
- Exam 2022 p. 11 - With weights w_k
- Exam 2018 p. 9 - Adding the found path to the restricted master problem

1.12 Dynamic networks

- Exam 2022 p. 7 - Shortest path with and rewriting problem, while using original solution

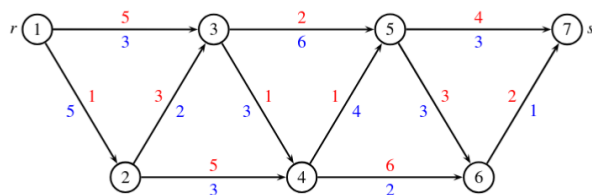


1.13 Constructing a network

Where they explain a problem and we have to model it. These will often/always be a network graphs.

- Exam 2022 p. 13 - From factories to costumers with lower bounds.
- Exam 2021 p. 13 - Projects with start, ends and man-hours needed

1.14 Resource constrained shortest path



node 1	node 2	node 3	node 4	node 5	node 6	node 7																																		
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- Exam 2021 p. 6 - Where the forward and backward runs are given